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Scope and Requirements

1. Purpose

This document sets out the scope, functional and non-functional requirements, delivery model, support obligations and contractual expectations for the design, build, integration, migration and ongoing maintenance of a Loan Management System (LMS) supporting Sharia-compliant (Murabaha-based) microfinancing products offered through the Bede mobile application.

Foo is expected to deliver the LMS and all integrations described herein, including integrations with the existing mobile application and mobile backend that Foo built.

All figures shown in square brackets are indicative and subject to agreement.

Background, Context, Scope Summary

2. Background and Context

- Bede is a fintech licensed by the Central Bank of Bahrain (CBB) under Rulebook Volume 5.
 - Financing is offered on a Murabaha basis; all products, calculations and accounting must be Sharia-compliant and subject to Sharia Supervisory Board approval.
 - Customers apply for and manage financing exclusively through the Bede mobile application. Staff operate the system through a back-office console (currently CMS).
 - Infrastructure is hosted in AWS and managed by Bede.
 - An in-house Eligibility Engine determines customer eligibility; the LMS must consume its output.
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3. Scope Summary

3.1 In scope

1. LMS platform: product configuration, loan lifecycle, repayments, collections, accounting sub-ledger, reporting data access.
2. All integrations listed in Section 5, including mobile app and mobile backend integration.
3. Back-office console for operations, finance, risk and compliance users.
4. Migration of existing customers and loans, including simulation cycles and cut-over.
5. Deployment packages, documentation, training and knowledge transfer.
6. Hypercare, warranty, 24x7 support and monthly security/hotfix releases.

3.2 Out of scope

- Provisioning and management of AWS infrastructure (Bede responsibility; Vendor provides specifications).
- The Eligibility Engine itself (in-house).

Functional Requirements

4.1 Product configuration

Requirement	Details
FR-01	Fully configurable Sharia-compliant financing products with no code changes: term, min/max amount, profit rate, fees, repayment frequency, grace periods, eligibility tags.
FR-02	Product variants driven by: type of good being financed, customer nationality, credit report outcome, and any other attribute exposed by the Eligibility Engine.
FR-03	Repayment schedule templates that are customisable and reusable, including irregular patterns (e.g., pay one month, skip two), bullet, step-up/step-down and seasonal schedules.
FR-04	Configurable repayment allocation order (fees / profit / principal), day-count convention, rounding rules, weekend and public-holiday handling.
FR-05	Product versioning: changes apply to new contracts only unless explicitly migrated; full history retained.
FR-06	Sharia Supervisory Board approval recorded against each product version as an acceptance gate before activation.

4.2 Murabaha mechanics and accounting

Requirement	Details
FR-07	Explicit Murabaha contract model: cost price, profit amount, total sale price; commodity purchase then sale to customer.

FR-08	Generation of the Murabaha sale contract and offer/acceptance documents in Arabic and English, presented and e-signed in the app before disbursement, stored with the loan record.
FR-09	Profit recognition rules configurable per product; no compounding of profit; late payment charges routed to a designated charity account, not to income.
FR-10	Profit rebate (ibra'') rules configurable per product for early and partial settlement.
FR-11	Full accounting sub-ledger: chart of accounts, automated journal postings for every financial event, month-end and year-end close, and audit-ready trial balance.
FR-12	IFRS 9 / AAOIFI staging and provisioning outputs (days past due, stage classification, expected credit loss inputs) per loan and portfolio.
FR-13	Daily automated reconciliation of the LMS sub-ledger against the Core Banking System with exception reporting.

4.3 Loan lifecycle

Requirement	Details
FR-14	Application: customer applies from the app; the LMS receives eligibility output and returns eligible products, terms and indicative schedules.
FR-15	Approval workflow with configurable stages, maker-checker, and automated approval where rules permit.
FR-16	Disbursement to customer account or merchant/commodity counterparty, with status tracked against Core Banking confirmation.
FR-17	Repayment via the app through integrated payment gateways; repayments received through Core Banking are also matched and applied.

FR-18	Early settlement with configurable rules per product: settle without profit, with profit, with a fixed fee, or with a percentage of the settled amount. Settlement quote generated in-app.
FR-19	Partial payment with the same configurable charging options as FR-18 and automatic schedule recalculation.
FR-20	Loan top-up with configurable eligibility, re-underwriting via the Eligibility Engine, and treatment as new contract or consolidated contract per product configuration.
FR-21	Multiple concurrent finances per customer, with portfolio-level exposure limits configurable per product and customer segment.
FR-22	Deferment for a configurable number of days or months, with configurable limits per product and treatment of profit during deferment.
FR-23	Restructuring and rescheduling (staff-initiated) with maker-checker approval and full audit trail.
FR-24	Write-off, fee waiver, refund and overpayment handling with maker-checker approval.
FR-25	Customer statements, repayment schedules and settlement quotes available on demand from the app and the back-office console.

4.4 Collections and defaults

Requirement	Details
FR-26	Default and delinquency rules configurable per product (days past due thresholds, amount thresholds, missed instalment count).
FR-27	Delinquency staging (e.g., pre-due reminder, 1–30, 31–60, 61–90, 90+, legal) with configurable stage names, thresholds and actions.

FR-28	Automated notifications to customers and to internal teams, with templates configurable per product, per stage and per channel.
FR-29	Notification channels: SMS, push, email and WhatsApp; Arabic and English; delivery status tracked.
FR-30	Collections work queues in the back-office console, promise-to-pay tracking and contact history.

4.5 Back-office console

Requirement	Details
FR-31	Role-based access control with SSO (SAML/OIDC) and MFA; segregation of duties enforced.
FR-32	Maker-checker on all configuration changes and financial adjustments.
FR-33	Immutable audit trail of every configuration change, data change and user action, exportable for Internal Audit and CBB.
FR-34	Customer 360 view: profile, contracts, schedules, payments, communications, documents.

4.6 Reporting and data access

Requirement	Details
FR-35	Analytics team access to readable, structured loan data with zero impact on the transactional system, via a read replica or change-data-capture feed to Bede data warehouse.
FR-36	Full data model documentation: entity-relationship diagram and data dictionary, maintained across releases.
FR-37	Structured extracts to support CBB regulatory returns and Sharia audit.

FR-38	Operational reports: portfolio, disbursements, collections, delinquency aging, settlements, profit recognised.
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4.7 Batch processing

Requirement	Details
FR-39	End-of-day and periodic jobs (accruals, aging, provisioning, dunning, schedule generation) that are monitored, idempotent, re-runnable and alert on failure.

Integration Requirements

#	System	Direction	Purpose
INT-01	Mobile application and existing mobile backend	Bi-directional	All customer journeys in Section 4.3; built by Foo
INT-02	Eligibility Engine (in-house)	Inbound to LMS	TBD Receive eligibility outcome; return eligible products and terms or Return all products to Eligibility Engine and Eligibility Engine returns eligible products to customer
INT-03	Commodity platform	Bi-directional	Commodity purchase and sale confirmation for Sharia compliance
INT-04	Payment gateways	Inbound	Repayment status, reconciliation, refunds

INT-05	Core Banking System	Bi-directional	Disbursement fund movement, repayment detection, daily reconciliation
INT-06	Credit bureau (Bahrain Credit Reference Bureau / BENEFIT)	Outbound	Reporting of facilities, delinquencies and closures
INT-07	AML / sanctions screening	Outbound	Screening hooks at onboarding and disbursement
INT-08	Notification providers (SMS, push, email, WhatsApp)	Outbound	Customer and internal notifications
INT-09	Company data warehouse	Outbound	Read replica / CDC feed (FR-35)
INT-10	Identity provider (SSO)	Inbound	Staff authentication

Integration standards (apply to all):

- OpenAPI 3.x specifications delivered and maintained for every API.
- Idempotent operations, retry with back-off, dead-letter handling, and automated reconciliation for every external system.
- Event/webhook support for status changes.
- API versioning with a documented deprecation policy.
- No customer PII in URLs, query strings or logs.

Non-Functional Requirements

6.1 Availability and performance

Requirement	Details
NFR-01 Availability	[99.9%] measured monthly, excluding agreed maintenance windows.
NFR-02 Recovery objectives	[1 hour]; Recovery Point Objective: [15 minutes].
NFR-03 Throughput & latency	[X] transactions per second; API response time p95 under [500 ms] for customer-facing calls.
NFR-04 Performance testing	Performance test at [3–5x] projected peak volume before go-live, with results delivered.
NFR-05 Deployment & DR	Multi-AZ deployment in AWS Ireland (eu-west-1); documented disaster recovery procedure tested before go-live and annually.

6.2 Security

Requirement	Details
NFR-06 Data residency	all customer data stored and processed in Bede's AWS account.
NFR-07 Encryption	Encryption at rest and in transit (TLS 1.2+); secrets held in a managed vault, never in code or configuration files.

NFR-08 PII logging	Field-level masking of PII in technical logs; a separate PII-bearing log tier accessible only to authorised Company staff.
NFR-09 Secure SDLC	SAST, DAST, dependency and container scanning in the CI/CD pipeline; results shared with Bede.
NFR-10 Penetration testing	Independent penetration test before go-live and annually thereafter, with all critical and high findings remediated before go-live.
NFR-11 Vulnerability SLAs	Critical [72 hours], High [14 days], Medium & Low [30 days - in the next security release].
NFR-12 Standards alignment	Alignment with OWASP ASVS and applicable CBB cybersecurity requirements.
NFR-13 Foo access	Foo personnel accessing any Company environment are named, background-checked and use individual, MFA-protected accounts.

6.3 Observability and operability

Requirement	Details
NFR-14 Telemetry	Structured logs, metrics, distributed traces and health endpoints exposed for Company monitoring tools.
NFR-15 Alerts & runbooks	Alert definitions and runbooks delivered for every critical failure mode.

NFR-16 Retention policies	Data retention and purge policies configurable to meet CBB and PDPL requirements.
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6.4 Architecture and deployment

Requirement	Details
NFR-17 Containerization & IaC	Containerized, deployable to Bede-managed Kubernetes (EKS); infrastructure and configuration as code (Terraform / Helm) allows instant deployment to recovery AWS regions.
NFR-18 Environment parity	Environment parity across dev, UAT and production.
NFR-19 Platform & upgrades	Software built on Apache Fineract or other similar open-source platform: upstream version pinned, documented upgrade path, and customizations delivered so as not to fork Bede off the upstream release line.
NFR-20 Deployment downtime	Zero-downtime deployments by default; any deployment requiring service interruption must be justified in writing and approved in advance by Bede.
NFR-21 Rollback support	Every deployment supports straightforward, tested rollback.

Delivery Model

7.1 Discovery

Item	Details
D-01	Foo deploys a team or expert on site to document detailed system specifications in workshops with Bede's business, technical and operational teams.
D-02	Deliverables: business requirements document, functional specification, solution architecture, integration specifications, data model, and a signed-off backlog with acceptance criteria per feature.

7.2 Environments and access

Item	Details
D-03	Bede provisions and manages all AWS environments.
D-04	Foo is granted deployment access to the development environment only.
D-05	UAT and production deployments are performed by Bede using Foo-supplied, tested deployment packages and step-by-step guides. Foo may be granted a view-only shadowing session during deployments.
D-06	Foo is granted access to PII-masked technical logs for troubleshooting (NFR-08). Access to logs containing PII remains with Bede.

7.3 Testing and acceptance

Item	Details
D-07	Definition of done and acceptance criteria agreed per feature before build.

D-08	Foo delivers unit, integration and system integration testing with evidence; supplies UAT test cases and anonymized test data.
D-09	Bede conducts UAT; acceptance is by written sign-off. Defects classified by severity; no Severity 1 or 2 defects open at go-live.
D-10	Performance (NFR-04), penetration (NFR-10) and disaster recovery (NFR-05) tests are acceptance gates.

7.4 Documentation, training and knowledge transfer

Item	Details
D-11	Administrator guide, operations runbooks, API documentation, deployment and rollback guides, architecture and data model documentation.
D-12	Training for operations, finance, risk, compliance and IT teams; recorded sessions and training materials retained by Bede.

Migration and Cut-over

Migration Requirements

Requirement	Details
M-01	Migration of all existing customers, contracts, schedules, repayment history, balances and documents.
M-02	Field-level data mapping document, agreed with Bede before the first simulation.
M-03	Reconciliation to the files on all balances and schedules, with a reconciliation report and Bede sign-off as the go-live gate.
M-04	Historical repayment history and original contract terms preserved and visible in the new system.
M-05	Multiple full simulation cycles of migration and cut-over using a new build of the app connected to the new LMS, without interrupting the existing app or backend.
M-06	Production cut-over downtime not to exceed [2 hours].
M-07	Documented and rehearsed rollback plan for cut-over.
M-08	Data freeze window, communication plan and post-migration validation checklist agreed in advance.

Hypercare, Warranty and Support

9. Hypercare, Warranty and Support

Service Levels

Service	Description
S-01 Hypercare	3–6 months post go-live with on-ground presence and daily review calls.
S-02 Warranty	5 years post go-live; all defects corrected at no cost.
S-03 24x7 support	24x7 support reachable by phone, email and ticketing portal, with a named escalation matrix up to executive level; support in English and Arabic.

S-04 Severity definitions and SLAs

Severity	Definition	Response	Resolution / workaround
1	Service down or financial integrity at risk	15 min	2 hours
2	Major function degraded, no workaround	30 min	8 hours
3	Minor function impaired, workaround exists	4 hours	3 business days
4	Cosmetic / low impact	1 business day	Next release

Monthly Service & Security Items

Item	Description
S-05	Monthly security and hotfix release, with a report covering all vulnerabilities and bugs identified, their severity, and remediation status.
S-06	Monthly service review covering SLA performance, incidents, root causes, capacity and upcoming changes.
S-07	Service credits / penalties for SLA breaches per Section 10.

Commercial and Contractual Terms / Governance / Assumptions

Commercial and Contractual Terms

Clause	Summary
C-01	Intellectual property: Bede owns all custom code, configuration, documentation and data. Source code held in a Bede-controlled repository or in escrow; inventory of all third-party and open-source licences delivered.
C-02	Payment: milestone-based, tied to written acceptance; retention of 20% released after 3 months post go-live.
C-03	Service credits: 5% of monthly support fees per SLA breach, capped at 20% per month; repeated breaches constitute grounds for termination.
C-04	Liquidated damages for delivery delay: 8% of contract value per week.
C-05	Change control: written change request process with impact assessment; rate card for out-of-scope work agreed at contract signature.
C-06	Key personnel named in the contract; replacement only with Bede approval. No subcontracting without prior written consent.
C-07	CBB outsourcing obligations: Bede and CBB right to audit; CBB access to Foo premises, staff and records; data location commitments; notification of material changes; compliance with CBB Rulebook Volume 5 outsourcing requirements.
C-08	Business continuity: Foo maintains and tests its own continuity plan covering delivery and support of this contract.

C-09	Exit and transition: on termination or expiry, Foo provides transition assistance for 6 months, full return of data and code, and destruction certificates for Bede data held.
C-10	Confidentiality (NDA) and a data processing agreement aligned to the Bahrain Personal Data Protection Law (PDPL).
C-11	Liability cap, indemnities (including IP infringement) and professional indemnity / cyber insurance at [amount].
C-12	Term: 5 years support and maintenance with annual renewal option and agreed pricing review mechanism.

Governance

Code	Governance Item
G-01	Steering committee (Bede executive sponsor, Head of IT, Foo account executive) meeting monthly.
G-02	Weekly project status report and meeting; shared RAID (risks, assumptions, issues, dependencies) log.
G-03	Single Foo project manager as point of accountability; single Bede counterpart.
G-04	All decisions, scope changes and acceptances recorded in writing.

Assumptions

- Bede provides timely access to subject matter experts, environments, test data and third-party sandbox credentials.
- Third-party systems (Core Banking, commodity platform, payment gateways, credit bureau) provide documented APIs and test environments.
- Sharia Supervisory Board reviews are scheduled by Bede; Foo supplies required product documentation.

Foo Response Requirements

Foo is requested to respond with:

Item	Description
1	Confirmation of compliance, partial compliance or non-compliance against every numbered requirement, with explanation.
2	Proposed solution architecture and platform (including any open-source or licensed components).
3	Delivery plan with milestones, resourcing and on-site commitments.
4	Commercial proposal covering implementation, support, and rate card.
5	Proposed SLA and service credit schedule.
6	Relevant references for Sharia-compliant lending implementations.